

Taghuchi Method Example

1) Experiment data (replicates) and per-run calculations

We use the smaller-is-better S/N formula:

$$S/N \text{ (dB)} = -10 \log_{10} \left(\frac{1}{n} \sum_{i=1}^n y_i^2 \right)$$

where n is number of replicates (here $n = 3$) and y_i are replicate measurements.

Runs (replicates), computed y_i^2 , mean of y^2 , S/N and arithmetic mean:

1. Replicates: 2.1, 2.0, 2.2

Squares: $2.1^2 = 4.41$, $2.0^2 = 4.00$, $2.2^2 = 4.84$

$\text{mean}(y^2) = (4.41 + 4.00 + 4.84)/3 = 13.25/3 = 4.4166666667$

$S/N = -10 \log_{10}(4.4166666667) = -6.450946 \text{ dB}$

Arithmetic mean $\bar{y} = (2.1 + 2.0 + 2.2)/3 = 2.100000$

2. Replicates: 1.9, 2.0, 1.8

Squares: 3.61, 4.00, 3.24

$\text{mean}(y^2) = (3.61 + 4.00 + 3.24)/3 = 10.85/3 = 3.6166666667$

$S/N = -10 \log_{10}(3.6166666667) = -5.583085 \text{ dB}$

$\bar{y} = 1.900000$

3. Replicates: 2.4, 2.3, 2.5

Squares: 5.76, 5.29, 6.25

$\text{mean}(y^2) = (5.76 + 5.29 + 6.25)/3 = 17.30/3 = 5.7666666667$

$S/N = -10 \log_{10}(5.7666666667) = -7.609248 \text{ dB}$

$\bar{y} = 2.400000$

4. Replicates: 1.7, 1.8, 1.6

Squares: 2.89, 3.24, 2.56

$\text{mean}(y^2) = (2.89 + 3.24 + 2.56)/3 = 8.69/3 = 2.8966666667$

$S/N = -10 \log_{10}(2.8966666667) = -4.618985 \text{ dB}$

$\bar{y} = 1.700000$

5. Replicates: 1.5, 1.6, 1.7

Squares: 2.25, 2.56, 2.89

$\text{mean}(y^2) = (2.25 + 2.56 + 2.89)/3 = 7.70/3 = 2.5666666667$

$S/N = -10 \log_{10}(2.5666666667) = -4.0975 \text{ dB}$

$\bar{y} = 1.600000$

Taghuchi Method Example

6. Replicates: 2.0, 2.1, 2.0

Squares: 4.00, 4.41, 4.00

$$\text{mean}(y^2) = (4.00 + 4.41 + 4.00)/3 = 12.41/3 = 4.1366666667$$

$$S/N = -10 \log_{10}(4.1366666667) = -6.166505 \text{ dB}$$

$$\bar{y} = 2.033333$$

7. Replicates: 1.8, 1.9, 1.7

Squares: 3.24, 3.61, 2.89

$$\text{mean}(y^2) = (3.24 + 3.61 + 2.89)/3 = 9.74/3 = 3.2466666667$$

$$S/N = -10 \log_{10}(3.2466666667) = -5.114377 \text{ dB}$$

$$\bar{y} = 1.800000$$

8. Replicates: 2.2, 2.1, 2.3

Squares: 4.84, 4.41, 5.29

$$\text{mean}(y^2) = (4.84 + 4.41 + 5.29)/3 = 14.54/3 = 4.8466666667$$

$$S/N = -10 \log_{10}(4.8466666667) = -6.854432 \text{ dB}$$

$$\bar{y} = 2.200000$$

9. Replicates: 1.6, 1.5, 1.6

Squares: 2.56, 2.25, 2.56

$$\text{mean}(y^2) = (2.56 + 2.25 + 2.56)/3 = 7.37/3 = 2.4566666667$$

$$S/N = -10 \log_{10}(2.4566666667) = -3.903462 \text{ dB}$$

$$\bar{y} = 1.566667$$

(These match the S/N values used earlier:

[−6.450946, −5.583085, −7.609248, −4.618985, −4.093695, −6.166505, −5.114377, −6.854432, −3.903462]
.)

2) Overall average S/N

$$\text{Overall S/N average} = \frac{1}{9} \sum_{i=1}^9 S/N_i = \frac{-6.450946 - 5.583085 - 7.609248 - 4.618985 - 4.093695 - 6.166505 - 5.114377 - 6.854432 - 3.903462}{9}$$

(rounded to 6 decimals above)

3) Average S/N by factor level (for each factor A, B, C, D)

We use the experimental L9 allocation (rows 1..9 correspond to runs). For each factor and each level (1,2,3) compute the mean of S/N values from the runs where that factor has that level.

Factor A (runs with A=1 are runs 1,2,3; A=2: runs 4,5,6; A=3: runs 7,8,9):

- A level 1: mean S/N = average of runs 1,2,3 = $(-6.450946 - 5.583085 - 7.609248)/3 = -6.5477597 \text{ dB}$
- A level 2: mean S/N = average of runs 4,5,6 = $(-4.618985 - 4.093695 - 6.166505)/3 = -4.9597283 \text{ dB}$
- A level 3: mean S/N = average of runs 7,8,9 = $(-5.114377 - 6.854432 - 3.903462)/3 = -5.2907570 \text{ dB}$

Taghuchi Method Example

Factor B (B=1 runs:1,4,7; B=2 runs:2,5,8; B=3 runs:3,6,9):

- B level 1: mean S/N = average runs 1,4,7 = $(-6.450946 - 4.618985 - 5.114377)/3 = -5.3947693$ dB
- B level 2: mean S/N = average runs 2,5,8 = $(-5.583085 - 4.093695 - 6.854432)/3 = -5.5104040$ dB
- B level 3: mean S/N = average runs 3,6,9 = $(-7.609248 - 6.166505 - 3.903462)/3 = -5.8930717$ dB

Factor C (C=1 runs:1,6,8; C=2 runs:2,4,9; C=3 runs:3,5,7):

- C level 1: mean S/N = average runs 1,6,8 = $(-6.450946 - 6.166505 - 6.854432)/3 = -6.4906277$ dB
- C level 2: mean S/N = average runs 2,4,9 = $(-5.583085 - 4.618985 - 3.903462)/3 = -4.7018440$ dB
- C level 3: mean S/N = average runs 3,5,7 = $(-7.609248 - 4.093695 - 5.114377)/3 = -5.6057733$ dB

Factor D (D=1 runs:1,5,8; D=2 runs:2,6,7; D=3 runs:3,4,9):

- D level 1: mean S/N = average runs 1,5,8 = $(-6.450946 - 4.093695 - 6.854432)/3 = -4.8160343$ dB
- D level 2: mean S/N = average runs 2,6,7 = $(-5.583085 - 6.166505 - 5.114377)/3 = -5.6213223$ dB
- D level 3: mean S/N = average runs 3,4,9 = $(-7.609248 - 4.618985 - 3.903462)/3 = -6.3608883$ dB

Choose the best level for each factor (for smaller-is-better we pick the largest S/N):

- A best = level 2 (-4.9597283 dB is largest)
- B best = level 1 (-5.3947693 dB)
- C best = level 2 (-4.7018440 dB)
- D best = level 1 (-4.8160343 dB)

4) Contribution (simple SS) calculation (Taguchi-style ranking)

We estimate contribution by computing the sum-of-squares of level effects for each factor:

$$SS_{\text{factor}} = \sum_{k=1}^3 n_k (\bar{\eta}_k - \bar{\eta})^2$$

where $\bar{\eta}_k$ = average S/N at level k , $\bar{\eta}$ = overall S/N average, and n_k is number of runs at that level (here $n_k = 3$ for all).

Overall S/N average $\bar{\eta} = -5.599415$ dB.

Taghuchi Method Example

Compute for each factor:

- A:
 - level1 diff = $-6.5477597 - (-5.599415) = -0.9483447 \rightarrow \text{squared} = 0.899389$
 - level2 diff = $-4.9597283 - (-5.599415) = +0.6396867 \rightarrow \text{squared} = 0.409195$
 - level3 diff = $-5.2907570 - (-5.599415) = +0.3086580 \rightarrow \text{squared} = 0.095571$
 - $SS_A = 3(0.899389 + 0.409195 + 0.095571) = 4.211479$ (approx)
- B:
 - level1 diff = $-5.3947693 - (-5.599415) = +0.2046457 \rightarrow \text{squared} = 0.041888$
 - level2 diff = $-5.5104040 - (-5.599415) = +0.0890110 \rightarrow \text{squared} = 0.007922$
 - level3 diff = $-5.8930717 - (-5.599415) = -0.2936567 \rightarrow \text{squared} = 0.086167$
 - $SS_B = 3(0.041888 + 0.007922 + 0.086167) = 0.408111$
- C:
 - level1 diff = $-6.4906277 - (-5.599415) = -0.8912127 \rightarrow \text{squared} = 0.794256$
 - level2 diff = $-4.7018440 - (-5.599415) = +0.8975710 \rightarrow \text{squared} = 0.805631$
 - level3 diff = $-5.6057733 - (-5.599415) = -0.0063583 \rightarrow \text{squared} = 0.000040$
 - $SS_C = 3(0.794256 + 0.805631 + 0.000040) = 4.799802$
- D:
 - level1 diff = $-4.8160343 - (-5.599415) = +0.7833807 \rightarrow \text{squared} = 0.613688$
 - level2 diff = $-5.6213223 - (-5.599415) = -0.0219073 \rightarrow \text{squared} = 0.000480$
 - level3 diff = $-6.3608883 - (-5.599415) = -0.7614733 \rightarrow \text{squared} = 0.579068$
 - $SS_D = 3(0.613688 + 0.000480 + 0.579068) = 3.582021$

Total SS = $SS_A + SS_B + SS_C + SS_D = 13.001413$ (approx).

Percentage contribution (approx):

- A: $100 \times 4.211479/13.001413 = 32.39\%$
- B: $100 \times 0.408111/13.001413 = 3.14\%$
- C: $100 \times 4.799802/13.001413 = 36.92\%$
- D: $100 \times 3.582021/13.001413 = 27.55\%$

So C ~ 36.9%, A ~ 32.4%, D ~ 27.6%, B ~ 3.1% (ranking of importance).

5) Predicted S/N at the optimal combination and conversion to predicted response

Chosen best levels (from §3): A_2, B_1, C_2, D_1 .

Taguchi predicted S/N at optimal combination = overall S/N average + sum of (level_mean - overall) for each chosen level:

$$\text{Predicted S/N} = \bar{\eta} + \sum_{\text{factors}} (\bar{\eta}_{\text{chosen level}} - \bar{\eta})$$

Taghuchi Method Example

Compute numeric:

- $\bar{\eta} = -5.599415$
- Contribution from A2: $-4.9597283 - (-5.599415) = +0.6396867$
- Contribution from B1: $-5.3947693 - (-5.599415) = +0.2046457$
- Contribution from C2: $-4.7018440 - (-5.599415) = +0.8975710$
- Contribution from D1: $-4.8160343 - (-5.599415) = +0.7833807$

Sum of contributions = $0.6396867 + 0.2046457 + 0.8975710 + 0.7833807 = 2.5252841$

Predicted S/N = $-5.599415 + 2.5252841 = -3.0741310$ dB

Convert predicted S/N back to an approximate predicted RMS response (for smaller-is-better S/N):

$$S/N = -10 \log_{10}(\overline{y^2}) \Rightarrow \overline{y^2} = 10^{-S/N/10}$$

Then predicted RMS = $\sqrt{\overline{y^2}}$.

Compute:

- $\overline{y^2} = 10^{-(-3.0741310)/10} = 10^{0.30741310} = 2.0296123647$
- Predicted RMS = $\sqrt{2.0296123647} = 1.4246446450$ (same units as y)

So **predicted RMS ≈ 1.42465 mm** (approx). That is the predicted typical magnitude of the response when using A2,B1,C2,D1 — you should run confirmation experiments at that combination to verify.